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New Lubricating Oil Filter System

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

This service/parts topic describes a redesign of the lubricating oil filtration system on the QST30 engine.

The old system consists of a full-flow filter head and a separate bypass filter head mounted in various locations on the engine. The full-flow filter head mounts four full-flow oil filters. The bypass filter head mounts two bypass oil filters. Filtered oil exiting the bypass filters is routed either to the number 6 right-bank cam follower cover or to the front cover where it drains to the oil pan.

The new lubricating oil filtration system was designed to utilize four Venturi Combo oil filters, Fleetguard® LF9001, Part Number 3406809. The LF9001 filters use patented technology to combine both full-flow and bypass filtration in a single filter. The new system consists of one Venturi Combo filter head, Part Number 3093596, that mounts four LF9001 filters. As a result, the bypass filter head and related components from the old system are **not** necessary.

The Venturi Combo filter head uses four replaceable adapters that are bolted to the head to provide the mounting point for the LF9001 filters. The old filter head uses adapters that are threaded into the head for filter mounting. It is **not** possible to mount the Venturi Combo adapters to the old full-flow filter head. However, the Venturi Combo filter head will mount directly in place of the old full-flow filter head. If this is done, the bypass filtration system can be removed.

The Venturi Combo lubricating oil filtration system offers several advantages over the old system. The primary advantage is the ability to extend oil and filter change intervals to 500 hours when Venturi Combo filters are used with premium grade engine oil. Premium grade oils meet API class CH4 specification, ACEA designation E5, and Cummins Engineering Standards 20071, 20072, and 20076. Valvoline® Premium Blue® oil meets all specifications to be classified as a premium grade oil. Use of Venturi Combo filters with standard grade oils requires an oil and filter change interval of 250 hours, or the use of the chart method with fuel consumption monitoring and oil sample analysis.

The LF9001 filter uses synthetic StrataPore™ media, which has the following benefits over cellulose media:

- Improved hot and cold flow
- More consistent performance
- Increased resistance to media rupture.

The LF9001 filter uses an improved seal design, which is required with extended oil change intervals. The structural strength of the filter has been increased. Finally, the new system reduces the number of filters that are changed, reducing associated disposal costs.

Note : This is a product improvement and is not subject to campaign.

- Order upfit kit, Part Number 4089094, which includes all parts necessary to convert an old system to the Venturi Combo system.
- Remove the old filter head and replace with the Venturi Combo filter head. Replace the two grommets, Part Number 3093019, and two o-rings, Part Number 3092438, between the oil supply/return tubes and the filter head. Replace the two o-rings, Part Number 3093809, between the oil supply/return tubes and the oil transfer connection. Refer to Procedure 007-015 in the Troubleshooting and Repair Manual, QST30 Series Engines, Bulletin 3666217.
- Remove the bypass filter head, filter head bracket, and bypass oil supply and return hoses.
- On power generation engines, plug the bypass oil supply line at the oil transfer connection and the oil return line to the number 6 right-bank cam follower cover with threaded plug, Part Number 3094340, and a sealing washer, Part Number 3094092.

Torque Value: 23 n•m [204 in-lb]

- On industrial engines, plug the return line to the front gear cover with a threaded plug and o-ring, Part Number 3089238. The oil supply port on the Venturi Combo filter head comes with plugs installed.

Torque Value: 27 n•m [239 in-lb]

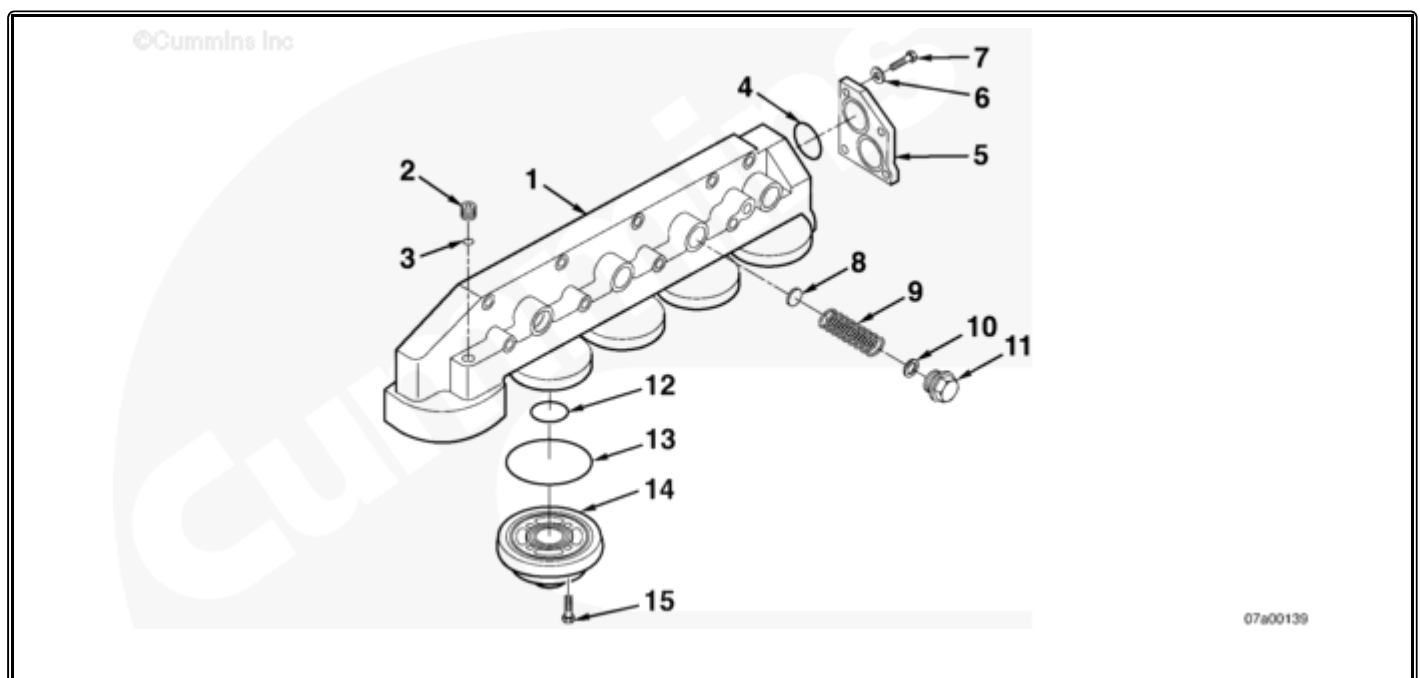


Table 1, Venturi Combo Lubricating Oil Filter Head, Part Number 3093596			
Item Number	Part Number	Description	Quantity
1	3093562	Lubricating filter head	1
2	3678924	Threaded plug	1
3	3678925	O-ring	1
4	3094129	O-ring (1 shown)	2
5	3094040	Cover plate (no ports)	1
6	3335076	Bevel washer (1 shown)	4
7	3093744	Hex head capscrew (1 shown)	4
8	200819	Bypass disc (1 shown)	4
9	3092078	Spring (1 shown)	4
10	3094836	Sealing washer (1 shown)	4
11	3094042	Threaded plug (1 shown)	4
12	3090126	O-ring (1 shown)	4
13	3090127	O-ring (1 shown)	4
14	3092967	Filter head adapter (1 shown)	4
15	3925402	Hex flange head capscrew (1 shown)	16
N/A*	3092582	Cover plate (3 ports)	1
N/A*	3089238	Threaded plug (Use with 3092582)	2
N/A*	3678921	Threaded plug (Use with 3092582)	1
*N/A indicates not applicable.			

